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Product Description

HSS/UDM

Version 11.000

Version Release Date: April 2026

Document Revision: 1

Related Release Version: 24.10.08

Distribution Status: Approved for Customer Distribution

Supplier Name: floLIVE Ltd.

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1. General Information

floLIVE's Connectivity Management Platform (CMP) is a cloud-based solution that encompasses a fully virtualized 3G/4G/LTE and 5G core network as well as a complementary suite of Business Support Systems (BSS) that enable our customers to take full ownership over their IoT business, be it device manufacturing and distribution or complete IoT connectivity management.

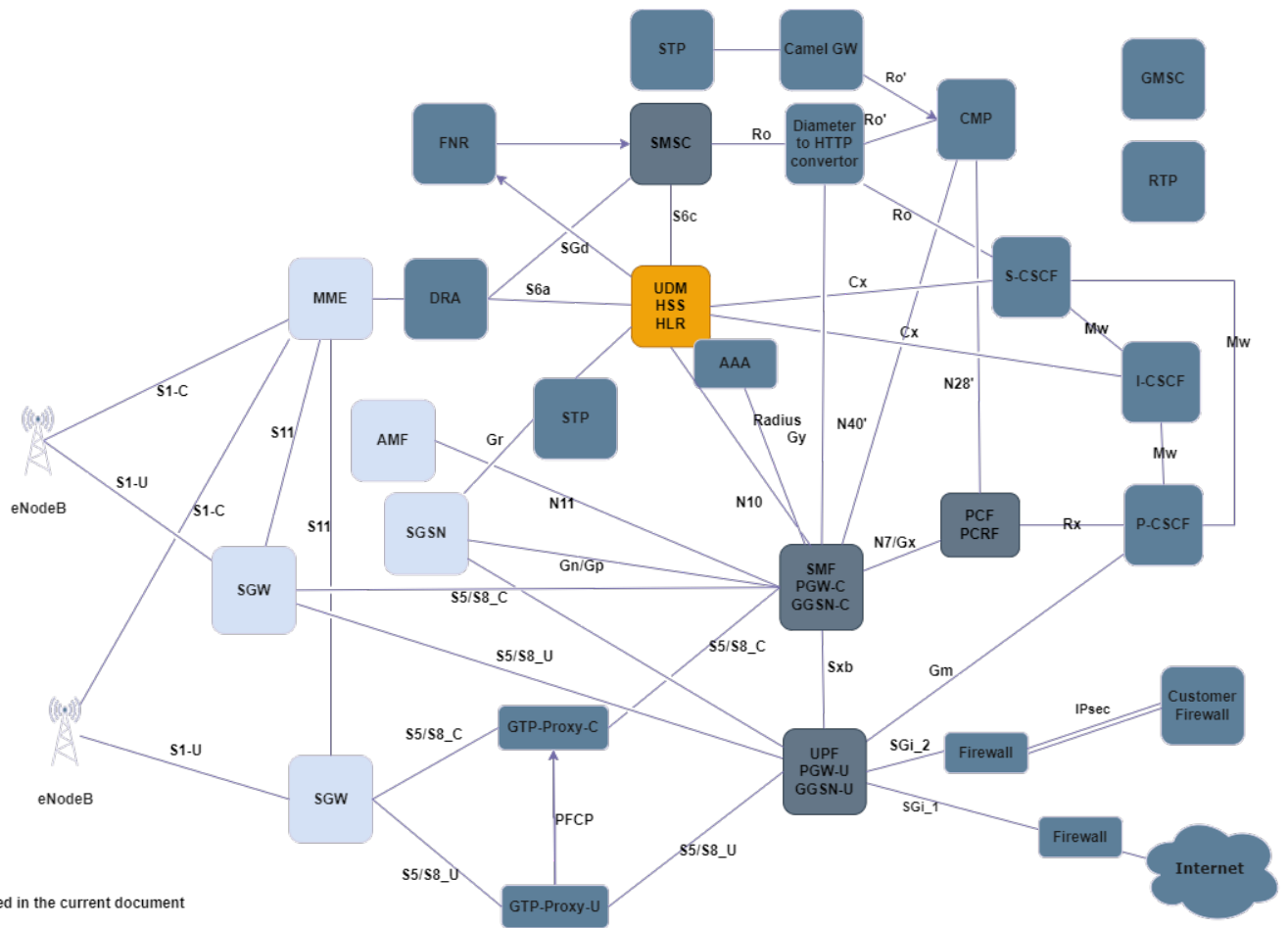
The platform provides the functionality necessary to support various workflows in managing the connectivity aspects of modern IoT and M2M device arrays.

floLIVE's CMP-as-a-Service solution comprises several subsystems and management applications.

The subsystems in use are the following:

- **floCONNECT Connectivity Management** - used for setting and managing accounts, users, SIMs, Self-care, Rules & Alerts, and more.
- **HLR/HSS** - Used for managing subscriber profiles, RAT Types, IMSI/MSISDN ranges and more.
- **GGSN/PGW (UPF & SMF)** - used for configuration and management of APNs, IP Pools, routing tables, etc.
- **SMSC** - used for configuration of SMS logic, forwarding rules and policies, external SMPP connections, and more.
- **floSIM Multi IMSI SIM Management** - used for configuration, management and control of SIM cards, business logic, PLMN definitions, OTA transactions, key management, and more.

Figure 1 - General Information



- Described in the current document
- Supported by floLIVE
- Not in the current portfolio

2. Overview

Each mobile network has its Home Location Registers (HLRs), Home Subscriber Servers (HSSs), or Unified Data Management (UDM).

The Home Location Register (HLR), Home Subscriber Server (HSS) is the main database element for subscriber information in a GSM/UMTS/LTE wireless network.

The UDM plays a role somewhat analogous to the HLR and HSS in 3G and 4G networks but with extended functionality that supports 5G. The UDM is responsible for managing user data, including subscriber authentication and service control. It holds the subscription data (user profiles) and supports network functions like authentication, service authorization, and location tracking.

This document provides an overview of floLIVE's HLR, HSS and UDM.

The HLR, HSS and UDM can be described as the master database for the subscribers of a given mobile network. It is the entity containing the subscription-related information to support the network entities handling mobility and user authentication and it is a key component of mobile networks such as GSM, UMTS, 4G/LTE, and 5G.

The HLR contains information about the subscriber's identity, his telephone number, the associated services, general subscriber location information, and any access restrictions for roaming.

The HSS contains users' subscription data such as the EPS-subscribed QoS profile and any access restrictions for roaming. It also holds information about the Packet Data Networks (PDN) to which the user can connect. This could be in the form of an Access Point Name (APN) (which is a label according to DNS naming conventions describing the PDN), or a UE address (indicating subscribed IP address(es)).

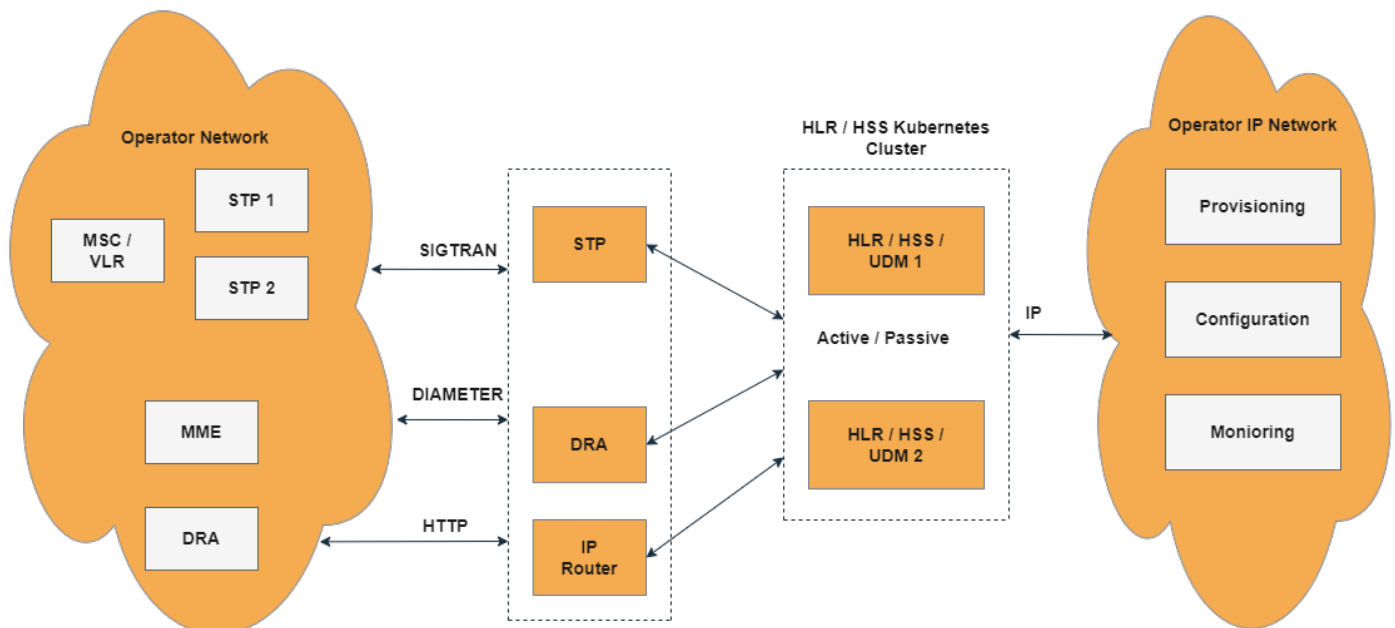
Unified Data Management (UDM) manages network user data in a single, centralized element. The technology is similar to the 4G network's home subscriber service (HSS). It is cloud based and designed for 5G specifically. In modern 5G systems UDM is a more advanced system that handles subscription data, authentication, and service management in a 5G environment.

3. Deployment Architecture

The diagram below depicts the HLR/HSS/UDM high-level architecture in an Active and Passive mode.

NOTE: The HLR/HSS/UDM can be installed on a cluster of Kubernetes and connected with floLIVE's SIGTRAN stack.

Figure 2 - HLR/HSS/UDM in a Kubernetes Cluster Mode



3.1. Kubernetes Deployment

The floLIVE HLR/HSS/UDM network element is deployed over Kubernetes.

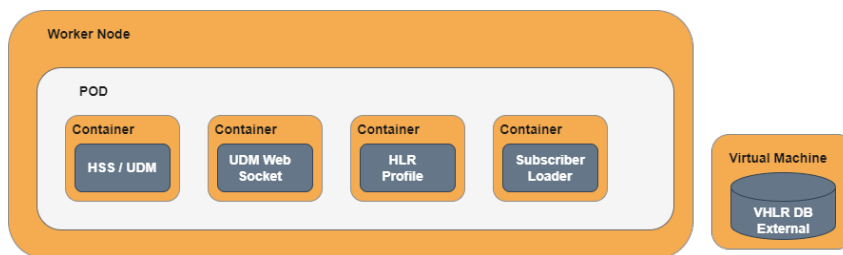
Kubernetes allows automatic scaling of the modules. In addition to scaling, Kubernetes aids with redundancies, performance and stability.

The HLR/HSS/UDM is installed in a docker that contains the required software components to function correctly. The docker installation also includes native redundancy and fail-safe options that keep the module working when a specific instance fails.

When HLR/HSS/UDM is deployed in Kubernetes, several machines operate pods that run the virtual function. The HLR/HSS/UDM operates within one of those pods, and the Kubernetes network maintains fidelity between the pods. This ensures that if one pod fails, due, for example, to machine malfunctions or connectivity issues, another pod can take the workload from the original pod.

The HLR/HSS/UDM dedicated database is maintained externally to the Kubernetes deployment.

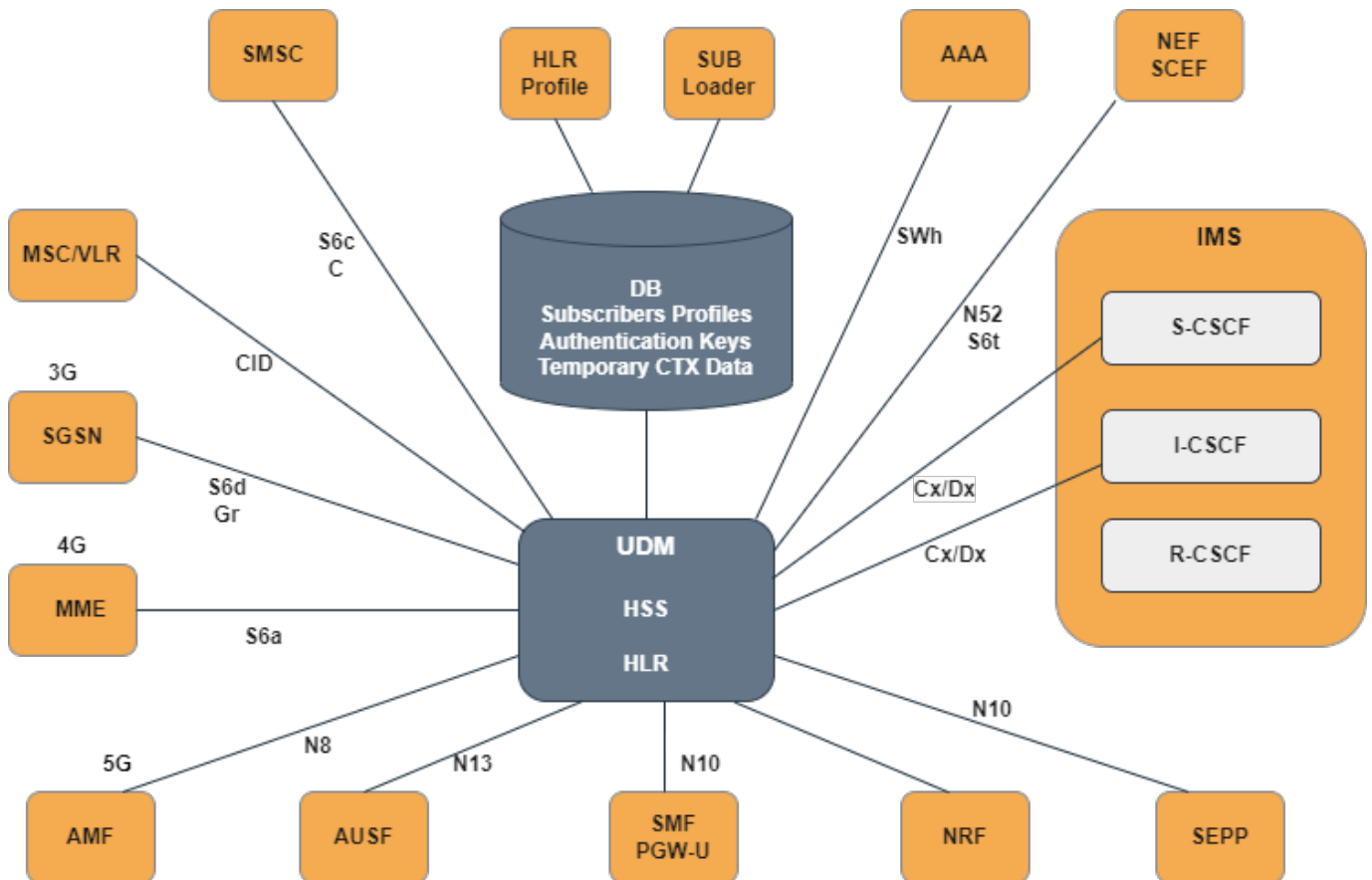
Figure 1 - HLR/HSS/UDM Kubernetes Deployment



4. Solution Architecture

The diagram below illustrate the HLR/HSS/UDM network architecture in the 3G, LTE (4G), and 5G networks, showing their relevant network interfaces:

Figure 1 - UDM Diagram



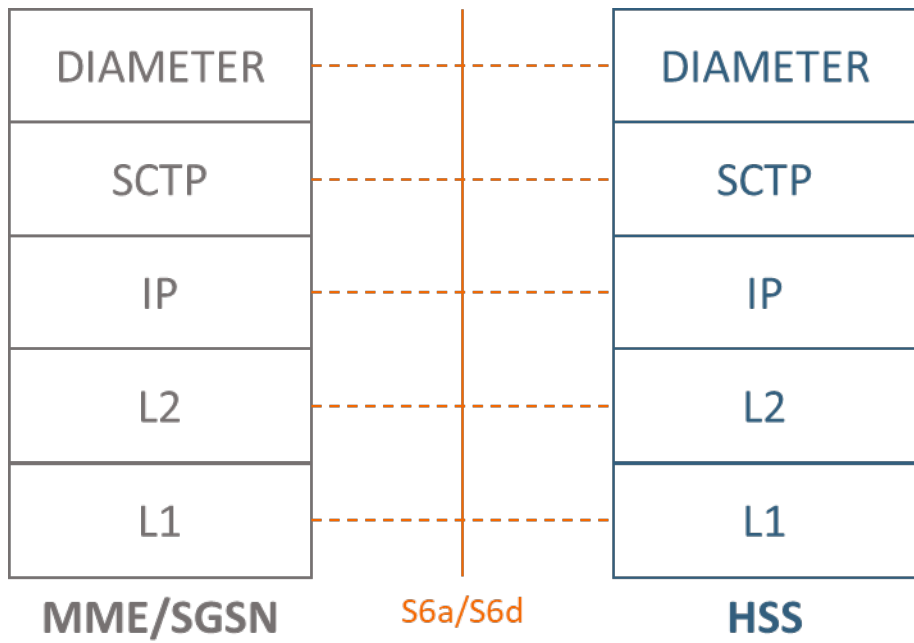
4.1. MME/SGSN Protocol Stack

The MME/SGSN protocol stack is used on both S6a/S6d interfaces and is based on the Diameter application.

The S6a/S6d Diameter application defines new commands and attribute-value pairs (AVPs).

The diameter messages over the S6a/S6d interfaces use the Stream Control Transmission Protocol (SCTP) as a transport protocol. The protocol stack is illustrated in the diagram below.

Figure 2 - Control Protocol Interface



5. SBI Protocol Stack

In the context of 5G networks, the User Data Management (UDM) function is responsible for managing user data, including subscription information, authentication, and authorization, across the system. It plays a critical role in the core network's architecture. HTTP/2 (HTTP Version 2) is a protocol that is used for efficient communication between network entities, particularly in service-oriented architectures like 5G. It brings improvements over HTTP/1.1, including multiplexing and better resource usage, which is especially important for modern network functions like UDM.

The protocol stack for 5G UDM HTTP/2 typically involves the following layers:

1. Application Layer (HTTP/2)

At the highest level, the communication between the 5G entities (like UDM and other network functions) happens via HTTP/2. It is designed to improve upon HTTP/1.1, particularly in mobile and 5G environments, by providing:

Multiplexing: Multiple HTTP requests and responses can be sent in parallel over a single TCP connection.

Header Compression: Reduces the overhead associated with HTTP headers, which is important for mobile and network function communication.

Stream Prioritization: Allows different streams to be prioritized, improving the performance of more critical services.

Role in UDM

UDM communicates over HTTP/2 to exchange information such as user profiles, authentication, authorization, and subscriber data with other network functions like the AMF (Access and Mobility Management Function) or the AUSF (Authentication Server Function).

HTTP/2 enables efficient, real-time interaction with low latency and minimal overhead, crucial for a smooth 5G experience.

2. Transport Layer (TCP)

HTTP/2 operates over Transmission Control Protocol (TCP) as its transport layer, which ensures reliable and ordered data delivery. The connection is established through a TCP handshake, and the data is transmitted over the established connection.

Role in UDM

TCP ensures that all critical user data exchanges (e.g., subscriber authentication, authorization, service requests) between the UDM and other core network functions are reliably transmitted.

3. Network Layer (IP)

The Internet Protocol (IP) is responsible for addressing and routing the data packets across the network, whether it's IPv4 or IPv6.

Role in UDM

IP addresses are used to route HTTP/2 messages between the UDM and other network elements. This allows UDM to securely and efficiently communicate over the global 5G network.

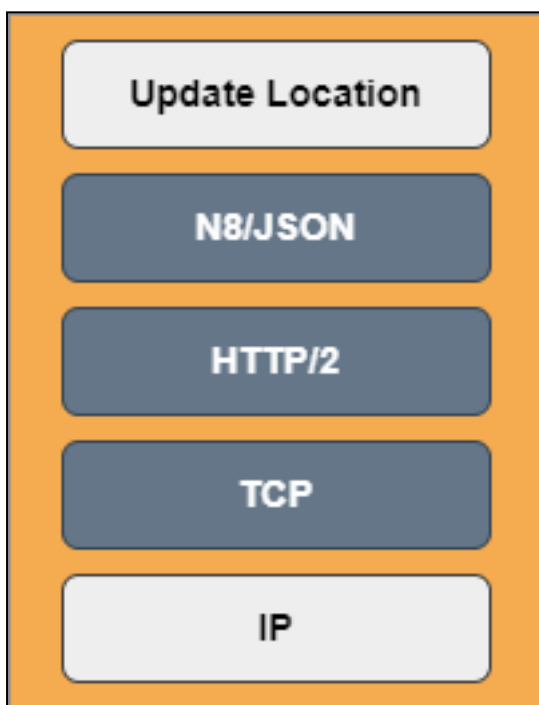
6. Service-Based Interface Protocol Stack

In the context of the SBI (Service-Based Interface) protocol stack used in 5G networks, the Update Location message is typically used for updating a subscriber's location within the network. The N8 reference point in the 5G architecture is used to connect the Access and Mobility Management Function (AMF) to the Unified Data Management (UDM).

The Update Location message is part of the communication between the AMF and the UDM, and it involves providing the updated location information for a particular subscriber. The message may be used when the mobile device (UE) moves to a new location or when the network needs to track its new location for service management.

You can review the example for the Update Location message in the SBI protocol stack with the N8 reference point.

Figure 3 - SBI Protocol Stack



7. Overload Management

The set of mechanisms that protect the UDM from becoming overwhelmed when it receives more requests than it can safely process is the overload management. Since the UDM is a critical 5G core function responsible for subscriber data, authentication, and policy information, overload can impact the entire network.

Overload management includes the following components:

- Startup delay
- Event number limitation
- Timeout checks

7.1. Startup Delay

A startup delay for session handling.

Nodes require time during startup to load scenarios and perform initialization before processing incoming events.

- A startup delay has been implemented to pause event processing and scenario execution temporarily during node startup.

The delay is configurable with a default of 10 seconds.

- Improve the management of `m_head_free_session`` to avoid errors related to handling free sessions in a multi-threaded environment.

7.2. Events Number Limitation

A limitation on the number of events that can occur per second.

Monitoring the number of events created for various interfaces, such as Diameter and Sigtran, and removing any events that exceed the limit.

The limit is configurable, with a minimum of 100 and a default of 1000.

7.3. Timeout Checks

A timeout check that determines the duration an event stays in the engine queue from its creation until it is processed.

Events are generated when a message is received from the network. Once created, the event waits in a queue until it is processed. If an event remains in the queue for too long, it should be discarded during a timeout check and not processed by the Base Band.

The timeout is configurable, default 10s, minimum 5s. If the value for some event is 0xFFFF, there is no timeout for it.

7.4. Overload Control

Core network nodes apply overload control by limiting the number of events accepted from the network interfaces and measuring the delay in event queue before event reaches processing engine.

7.4.1. Events Count

The nodes count the incoming messages on network interfaces and discard those exceeding the limit. This limit is defined as the number of messages accepted per second.

7.4.2. Queue Timeout

Received events are added to a queue waiting for a free engine (thread) to process them. If events remain in the queue for longer than a specified timeout, they are discarded. The configuration value for the timeout is set in seconds.

7.4.3. Events Under Control

The table lists the events that are monitored and the config values intbl_configurations that define the limits:

	Selection	Attribute	Default	Description
1	AAA	ARR_TIMEOUT	3	Radius Access Request Receive timeout
2	AAA	ARR_COUNT_LIMIT	500	Radius Access Request Receive, min. 100
3	AAA	ACR_TIMEOUT	3	Radius Accounting Request Receive timeout
4	AAA	ACR_COUNT_LIMIT	500	Radius Accounting Request Receive, min. 100
5	HLR	SAI_TIMEOUT not configurable in 24.10	3	GSM-MAP Send Auth Info timeout
6	HLR	SAI_COUNT_LIMIT	500	GSM-MAP Send Auth Info minimal applied value is 100
7	HLR	UL_CS_COUNT_LIMIT	500	GSM-MAP Update Location CS minimal applied value is 100
8	HLR	UL_PS_COUNT_LIMIT	500	GSM-MAP Update Location PS minimal applied value is 100
9	HLR	UL_TIMEOUT not configurable in 24.10	3	GSM-MAP Update Location timeout
10	HSS	AIR_TIMEOUT not configurable in 24.10	3	Diameter Auth Info Requests timeout
11	HSS	AIR_COUNT_LIMIT	500	Diameter Auth Info Requests minimal applied value is 100
12	HSS	UL_COUNT_LIMIT	500	Diameter Update Location Requests

	Selection	Attribute	Default	Description
				minimal applied value is 100
13	HSS	UL_TIMEOUT not configurable in 24.10	3	Diameter Update Location Requests timeout
14	PGW	CSR_TIMEOUT from 25.11	2	GTPC Create Session Request timeout
15	PGW	MBR_TIMEOUT from 25.11	3	GTPC Modify Bearer Request timeout
16	PGW	MBC_TIMEOUT from 25.11	3	GTPC Modify Bearer Command timeout
17	PGW	DMR_TIMEOUT	3	Disconnect Message Receive timeout
18	PGW	DMR_COUNT_LIMIT		Radius Disconnect Receive, min. 100
19	AAA	ARR_TIMEOUT	3	Radius Access Request Receive timeout
20	AAA	ARR_COUNT_LIMIT	500	Radius Access Request Receive, min. 100
21	AAA	ACR_TIMEOUT	3	Radius Accounting Request Receive timeout
22	AAA	ACR_COUNT_LIMIT	500	Radius Accounting Request Receive, min. 100

7.5. Signaling Limitations

SIGTRAN Limits (2G/3G):

- 500 SAI (Send Authentication Info) procedures per second
- 500 CS Update Location procedures per second
- 500 PS Update Location procedures per second

DIAMETER Limits (4G):

- 500 AIR (Authentication Information Request) procedures per second
- 500 ULR (Update Location Request) procedures per second
- 500 Notify procedures per second

8. HLR/HSS/UDM Application Features

8.1. Authentication Center (AuC)

The Authentication Centre (AuC) is the part of the HLR/HSS/UDM responsible for authenticating each SIM card that attempts to connect to the GSM/UMTS/LTE core. Once authentication is successful, the HLR/HSS/UDM can manage the SIM and related services. The AuC generates an Authentication Key (“Ki”), and its purpose is to encrypt all wireless communication between the MS and the core network. If the authentication process fails, no services are enabled for the specific SIM/Mobile Operator.

NOTE: The AuC uses a floLIVE Subloader module for uploading and provisioning new subscribers into the AuC module.

For SIM authentication, we support EPS-AKA protocol.

8.2. AuC Authentication Process

AuC supports these authentication methods:

- Comp 128-1/2 and Comp 128-4 (Milenage) type authentication
- The subscriber profiles are with 16 bytes OPC
- TUAK algorithm for encryption and authentication with 16 bytes TOPC

The authentication flow goes through the following main steps:

1. The subscriber requests access to the network, using identity information stored in the Security Identity Module (SIM) card.
2. The Serving GPRS Support Node (SGSN)/VLR forwards the request to the appropriate Authentication Center, hosted on the HLR/HSS.
3. The HLR/HSS returns a random 128-bit number (called “challenge”) and corresponding 32-bit “challenge response” (computed using the GSM A3 algorithm) to the SGSN/VLR.

NOTE: The SGSN/VLR temporarily stores the expected “challenge-response”. The shared secret key is not transmitted to the SGSN or the MS.

4. The MS computes the 32-bit response based on the 128-bit random number and its subscriber authentication key, again using the A3 algorithm.

NOTE: The authentication key is stored on both the SIM card and the HLR/HSS AuC.

5. The SGSN/VLR compares the response provided by the MS with that given by the HLR/HSS - If they match, the MS is successfully authenticated.

For UMTS connections, encryption occurs between the UMTS device and the Radio Network Controller (RNC).

8.3. 5G Authentication and Authorization

The 5G system meets the following requirements:

- **Subscription authentication:** The serving network authenticates the Subscription Permanent Identifier (SUPI) in the process of authentication and key agreement between UE and network.
- **Serving network authentication:** The UE authenticates the serving network identifier through implicit key authentication.

NOTE: The meaning of 'implicit key authentication' here is that authentication is provided through the successful use of keys resulting from authentication and key agreement in subsequent procedures.

NOTE: The preceding requirement does not imply that the UE authenticates a particular entity, e.g. an AMF, within a serving network.

- **UE authorization:** The serving network authorizes the UE through the subscription profile obtained from the home network. UE authorization is based on the authenticated SUPI.
 - **Serving network authorization by the home network:** Assurance is provided to the UE that it is connected to a serving network that is authorized by the home network to provide services to the UE. This authorization is 'implicit' in the sense that it is implied by a successful authentication and key agreement run.
 - **Access network authorization:** Assurance is provided to the user equipment (UE) that it is connected to an access network authorized by the serving network to provide services to the UE. This authorization is 'implicit' in the sense that it is implied by a successful establishment of access network security. This access network authorization applies to all types of access networks.
-

NOTE: The authentication method is 5G AKA.

8.3.1. Security Entities in 5G Core Network

The 5G System architecture introduces the following security entities in the 5G Core network:

- **AUSF:** Authentication Server Function;
- **ARPF:** Authentication credential Repository and Processing Function;
- **SIDF:** Subscription Identifier De-concealing Function;
- **SEAF:** Security Anchor Function.

8.4. AuC Security Features

The AuC has the following security features:

- The floLIVE AuC can communicate and integrate with HSMs (Hardware Security Module) external to the floLIVE system.

- Security methods for uploading and provisioning new subscribers into the AuC module:
 - Encrypted file encapsulating encrypted keys.
 - Secured API (HTTPS, PKCS #11).

8.5. HLR/HSS/UDM Location Information

The main function of the HLR/HSS/UDM is to manage the movement of the SIMs and phones. The following procedures have been implemented:

- Managing the mobility of subscribers through updating their position in administrative areas called 'Location Areas (LA)'; these are identified with a Location Area Code (LAC). The action of a user moving from one LA to another is followed by the HLR/HSS/UDM with a Location area update procedure.
- Sending subscriber data to a VLR or SGSN when a subscriber first roams there.
- Brokering between the G-MSC or SMSC and the subscriber's current VLR to allow incoming calls or text messages to be delivered.
- Removing subscriber data from the previous VLR when a subscriber has roamed away from it.
- Responsible for all SRI-related queries (that is, for invoking SRI, HLR/HSS/UDM reply with an SRI).

The HLR/HSS/UDM call flows are illustrated in the following flow diagrams:

- Location update.
- GPRS location update.
- LTE location update.

Figure 1 - HLR/HSS Location Update Call Flow

HLR/HSS Call Flow: Location Update

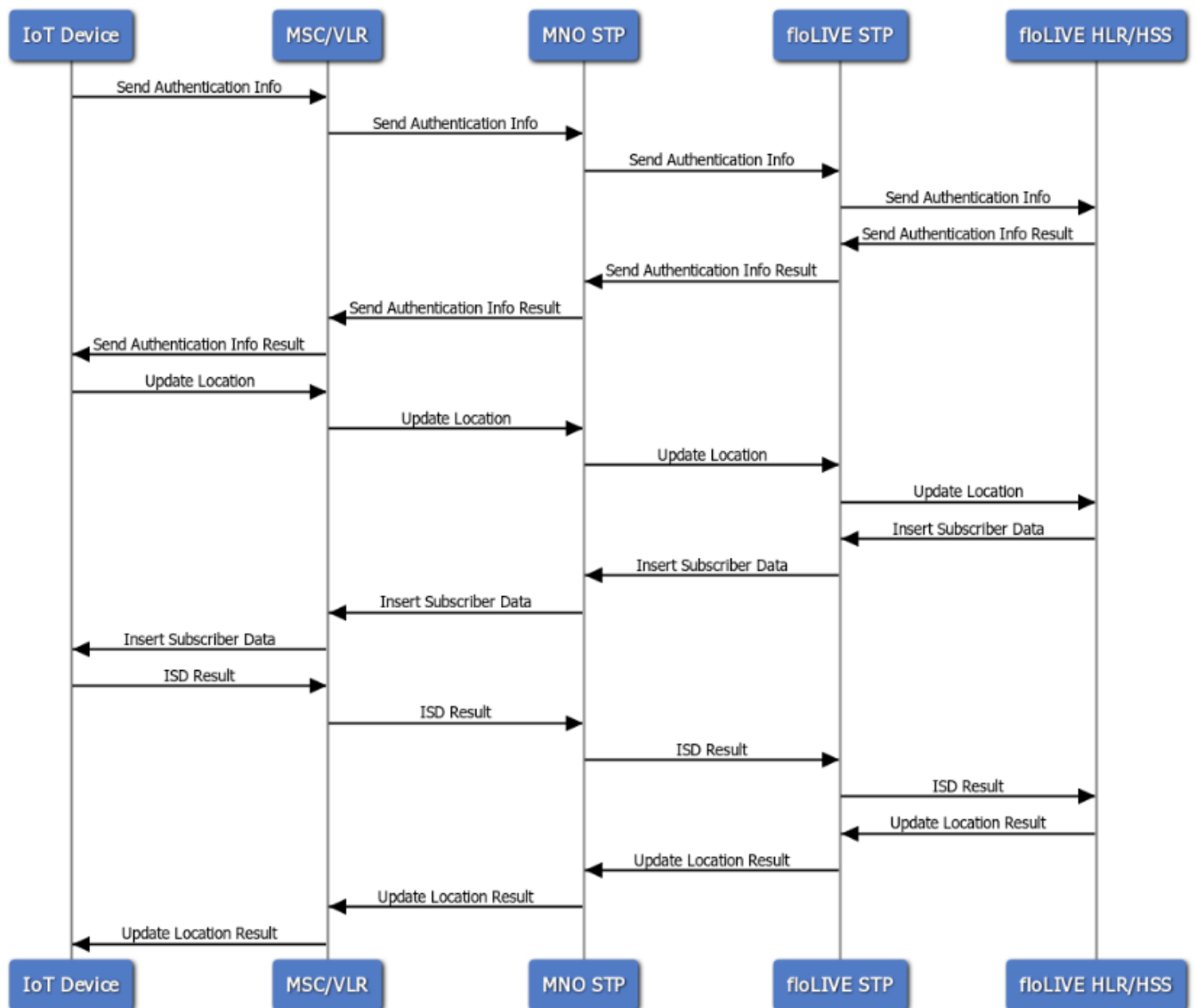


Figure 2 - HLR/HSS GPRS Location Update Call Flow

HLR/HSS Call Flow: GPRS Location Update

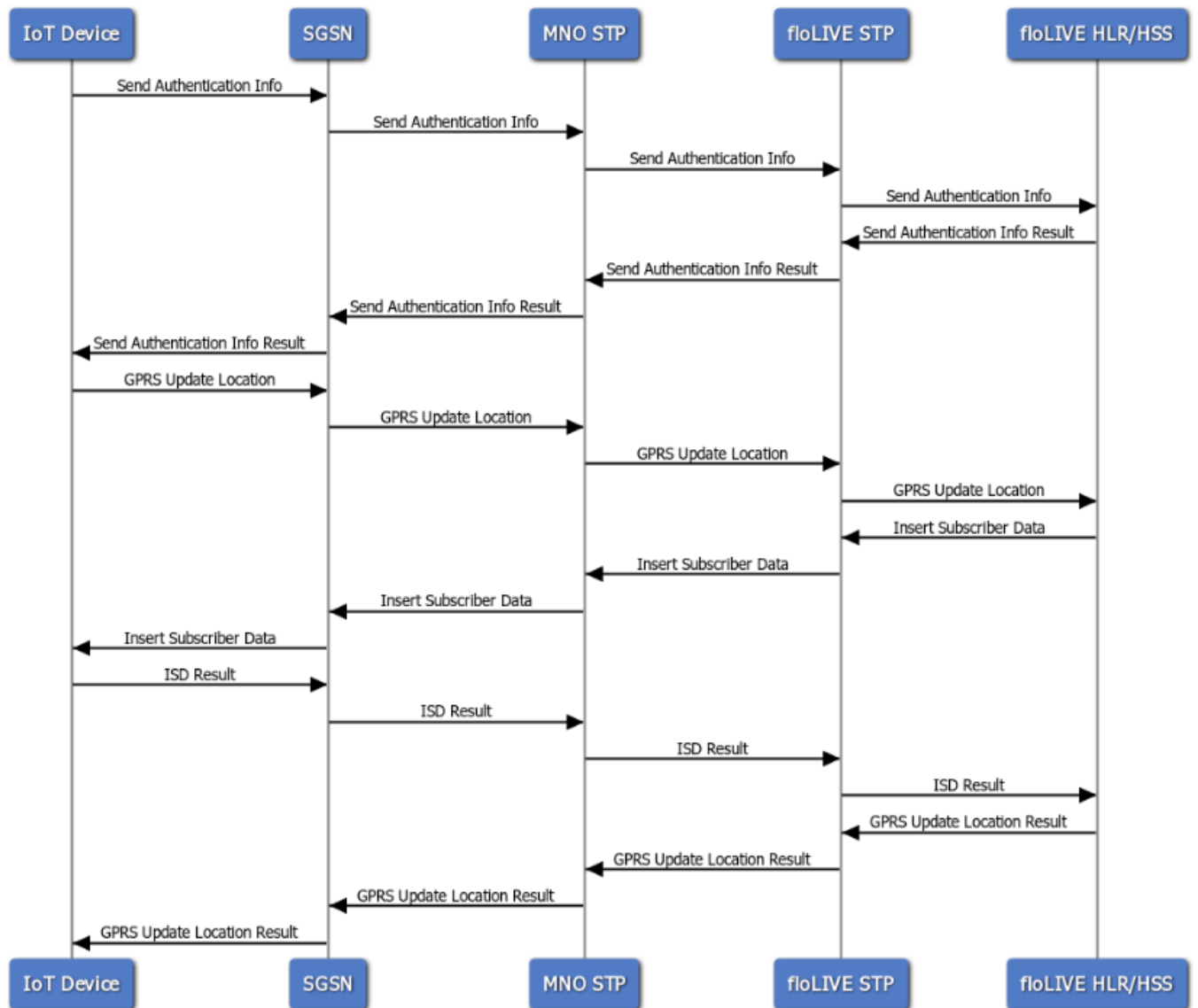
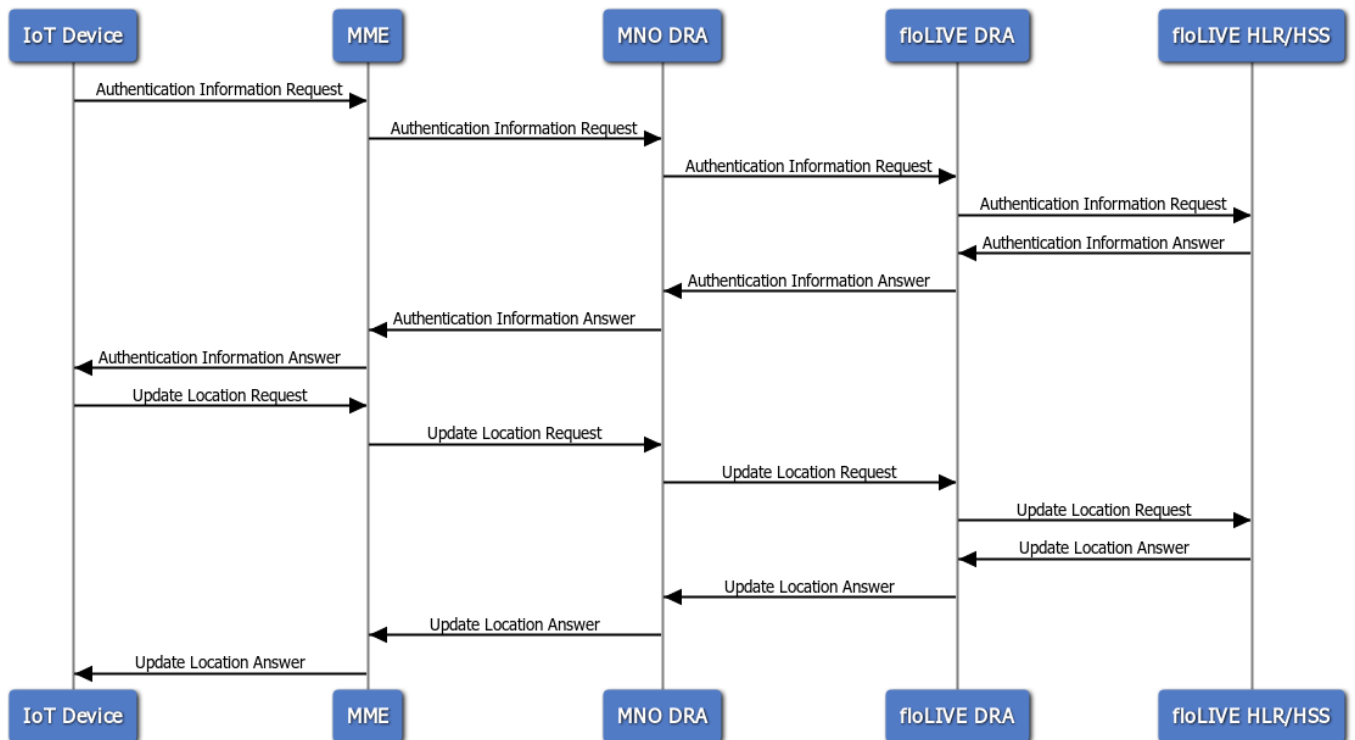


Figure 3 - HLR/HSS LTE Location Update Call Flow

HLR/HSS Call Flow: LTE Location Update



8.6. Mobility Management (MM)

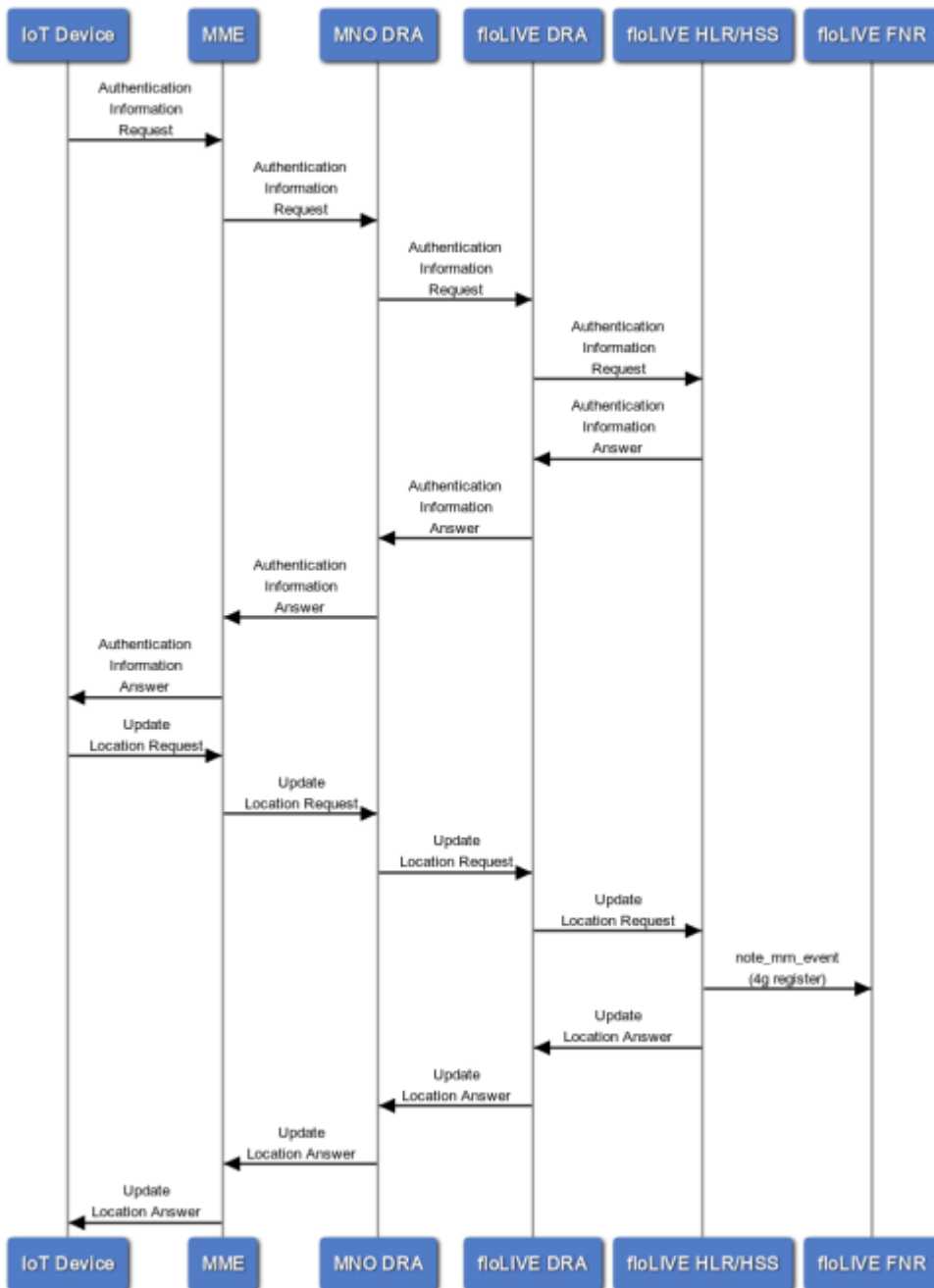
The Mobility Management (MM) interface provides a command event API interconnecting external systems or core modules. The interface allows notifying external systems of a successful termination of mobile network events such as Update location, Cancel Location, Purge MS, etc.

The MM functionality: is used for supporting multi-IMSI SIMs and can also be used for third parties that need mobility management information about a device.

The following flow diagram shows how the HLR/HSS updates the FNR for all subscribers' MM activity.

Figure 4 - HLR/HSS Mobility Management – LTE Register Event

HLR/HSS Mobility Management – LTE Register Event



9. SMS Mobile Terminated Barring Using Charging Characteristics Provisioning

In addition to barring SMS services through the subscriber profile, the floLive HSS supports SMS Mobile Terminated (MT) barring based on the subscriber's Charging Characteristics (CC) value, applied through the SRI-for-SM response.

Operators can configure the parameter "CC_PRE_PAID" under the "HLR" section in the tbl config. When this feature is enabled, the HSS performs a bitwise AND operation between the subscriber's CC value and the configured "CC_PRE_PAID" bitmask i.e. *if (CC & CC_PRE_PAID) == CC_PRE_PAID*.

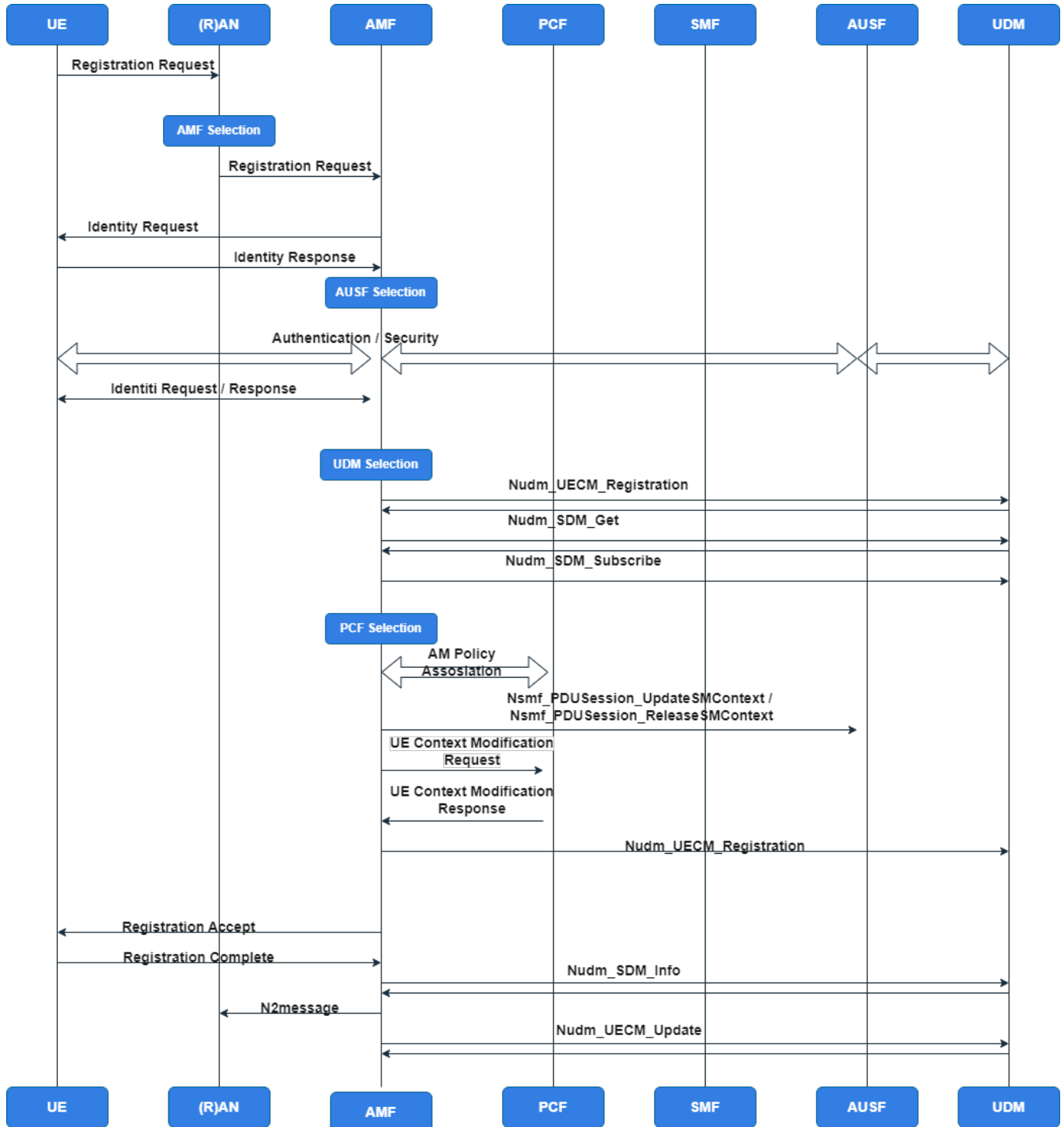
If the result of this operation equals the configured CC_PRE_PAID value, the HSS determines that the subscriber should not receive SMS MT messages. Consequently, the HSS returns an SRI-for-SM response with error cause 21 – "Facility Not Supported", indicating that the SMS delivery is rejected.

To activate this feature, the parameter CC_PRE_PAID must be configured with a value greater than 0 and less than 16.

10. Registration Procedure

Review the 5G registration procedure on the diagram below.

Figure 5 - Registration Procedure



10.1. Roaming

Each mobile network has its Home Location Registers (HLRs)/Home Subscriber Servers (HSSs) and Visitor Location Registers (VLRs). When a Mobile Switching Center (MSC) detects a mobile user's presence in the area covered by its network, it first checks the database to determine if the user is in their home area or if they are roaming, meaning they are a visitor.

- **User in Home Area:** HLR/HSS/UDM has the necessary information for initiating, terminating or receiving a call.
- **User is Roaming:** VLR contacts the user's HLR/HSS/UDM to get the necessary information to set up a temporary user profile.

The user's location is recorded in the HLR/HSS/UDM, and in case the user is roaming, it is also recorded in the VLR.

When a subscriber initiates a call:

- **User in Home Area:** MSC contacts the HLR/HSS prior to setting up the call
- **User is Roaming:** MSC contacts the VLR prior to setting up the call

When a subscriber receives a call (call routed to the home MSC):

- **User in Home Area:** Home MSC delivers the call immediately
- **User is Roaming:** Home MSC contacts the VLR to determine the appropriate switch in the roaming area to handle the arriving call and then transfers the call to the roaming area MSC

There can be several HLRs/HSSs per Public Land Mobile Network (PLMN), however the IMSI/MSISDN pair can be associated with only one logical HLR/HSS.

The HLR/HSS holds the details for every SIM card issued by the mobile phone operator. Each SIM has a unique identifier called IMSI, which is a primary key for each HLR/HSS record. MSISDN is the telephone number used for originating and terminating phone calls.

The primary MSISDN is used for initiating and terminating voice calls and SMS messages, however there are some cases in which a SIM has a secondary MSISDN associated with it, dedicated to fax and data.

10.2. Commercial Cloud Services

floLIVE's HLR/HSS can run on any commercial cloud service, such as:

- Amazon AWS,
- Microsoft Azure,
- Google and others.

Element	Description
	Queues data requests for later execution in the Command Registry The node can be configured remotely through a configuration database.
floLIVE Core	Core network elements communicate with UE in the network to gather data from them in response to LEA requests: Sends the gathered data to the Data Cache over TCP/IP over the X2/X3 protocol TCP/IP security is ensured using GnuTLS implementation
Data Cache	Gathers and dispatches data to the floLIVE Lawful Interception Node over TCP/IP; it sends the data over the X2/X3 protocol to the LEA

10.4. Steering Analysis

The floCORE HLR/HSS/UDM supports the steering functionality, which blocks or allows devices to register on MNO networks. The HLR/HSS/UDM supports the steering functionality on 3G, LTE/4G and 5G networks.

The HLR/HSS/UDM supports two steering methods:

- Permanent steering: Network steering for a single IMSI or a range of IMSIs, based on the SCCP address (GT prefixes).
- Steering by number of attempts: A counter keeps track of the total number of registered attempts for each GT; Steering is executed when the number of registered attempts reaches a threshold.

The HLR/HSS supports subscriber steering inside the subscriber's allowed networks, which are configured in the subscriber profile.

Steering Tree 3G/4G/5G

The system supports defining different steering trees per network type, 3G/ 4G or 5G. The Steering 3G/4G/5G decision tree identifies each subscriber with a Steering ID unique to that subscriber. The Steering 3G/4G/5G decision tree uses the Steering ID to identify the networks that are either allowing or blocking each subscriber.

11. Management

11.1. HLR/HSS/UDM Administration Tool

The HLR/HSS/UDM support web-based graphical user interface tools to manage the following HLR/HSS/UDM aspects:

- Profile management: Based on templates, categories, or individual settings
- Manage device bearer services: For the different types of communication services
- Manage device supplementary services: Name identity, forwarding settings, and barring
- Manage Data services settings: APNs, GPRS
- Steering analysis
- SMS settings
- Voice Settings
- HLR/HSS/UDM configuration settings

The HLR/HSS/UDM management tool supports an advanced searching tool that enables an easy and quick way to troubleshoot and edit device settings in real-time.

The screen capture below depicts the HLR/HSS/UDM web management tool:

Figure 7 - LR HSS Web Admin



11.2. Management APIs

The floLIVE solution supports REST APIs for managing and configuring the HLR/HSS network element.

floLIVE REST APIs provision all network elements; settings include pool management, routing and APN settings, number of bearers, search bearer by IMSI, bearer management (block, resume, delete), and more.

The HLR/HSS/UDM has Management and Operations APIs, such as:

- Cancel Location
- Send USSD
- Send USSN/USSR
- Send MSRN
- Suspend/Activate subscriber, and more.

NOTE: If an APN is missing for some reason, a message in the PCAP file indicates it.

11.3. HLR/HSS/UDM Monitoring

The HLR/HSS/UDM nodes support integration with a centralized monitoring system that continuously scans all HLR/HSS/UDM resources and notifies system and network administrators upon outages, malfunctions, disconnections, and more.

Periodic, as well as on-demand performance reports, are generated to enable quick and efficient drill-down and root-cause analysis.

The HLR/HSS/UDM supports both the SNMPs and Traps methods:

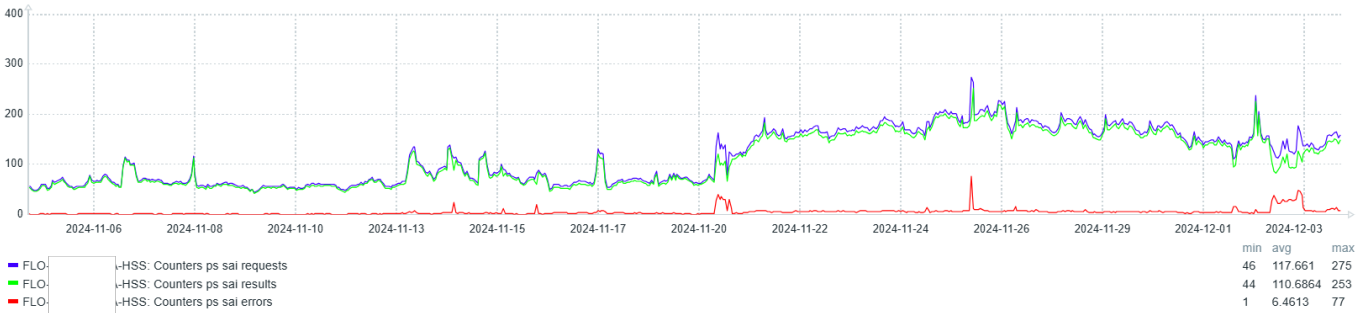
- Server and service resources:
 - CPU
 - Memory
 - Storage
- Service status:
 - Up/Down/Restart
 - SIGTRAN links status
 - Traffic threshold alarms
- Statistics:
 - Open sessions
 - Active resources
 - Hardware: Temperature, fans, power supply, and more.

NOTE: Traps are sent both upon a system event and periodically (for example, every 10 seconds).

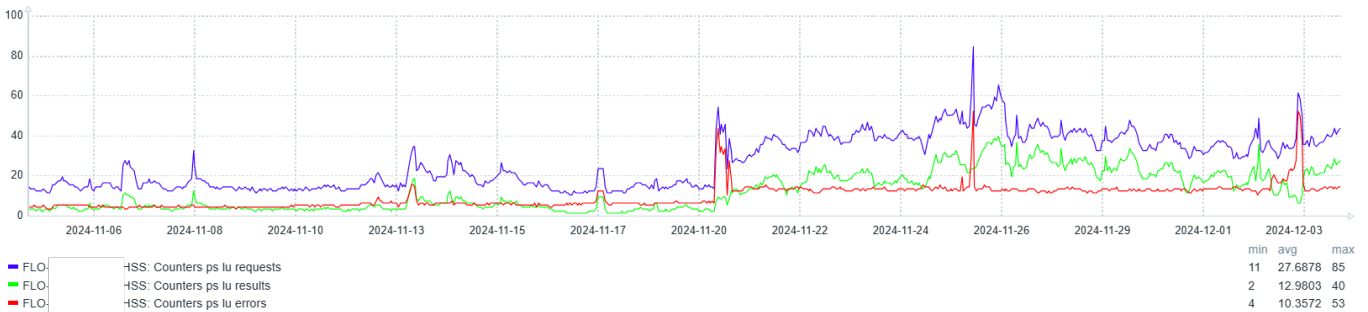
The screen capture below depicts a typical monitoring screen:

Figure 8 - Centralized Monitoring

3G PS Send Authentication Info (SAI)



3G PS Update Location (UL)



11.4. Alarms

The floLIVE core network is designed and implemented as a Telco grade system; the system generates and manages system alarms on all system layers and modules.

Alarms are generated by all core network nodes in real time and can be sent to any internal or external monitoring system via standard SNMP Traps.

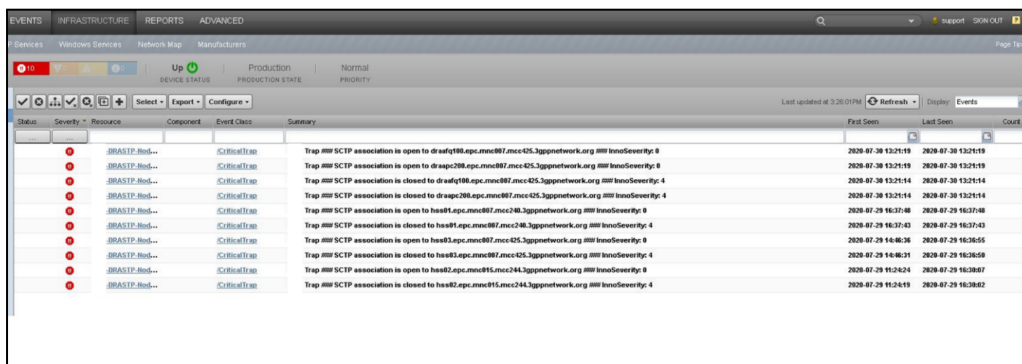
The system supports different types of alarms for different purposes, such as debugging, information, and critical alarm indications, and is based on alarm severity definitions.

System alarms can either send a binary indication, such as "Alarm Raised"/"Alarm Cleared", or a numeric value that is treated according to the alarm value for example.

EXAMPLE: Val<60% - Info, Val>80% - Critical.

This is a standard monitoring screen that displays Alarms:

Figure 9 - Typical Alarms Monitoring Screen



12. KPIs

The floLIVE core network supports KPI counters for the different core nodes.

System KPIs are a very important part of any monitoring system and allow an easy way to monitor the floLIVE core network system usage, picks, and trends.

KPIs are implemented as SNMP counters that can be read at any time by any standard monitoring system.

The following table displays a sample of SNMP counters that are supported by the HLR/HSS module:

KPI	Description	Reference
sri sm	Counter for "Send Routing Information" for SMS. Counter format is "Requests Results Errors". Example: 10 8 2	1.3.6.1.4.1.35150.1.2.3.10.1.2.2
sri (calls)	Counter for "Send Routing Information" for voice calls. Counter format is "Requests Results Errors". Example: 10 8 2	1.3.6.1.4.1.35150.1.2.3.11.1.2.2
mo ussr	Counter for "Mobile Originating" SMS. Counter format is "Requests Results Errors". Example: 10 8 2	1.3.6.1.4.1.35150.1.2.3.13.1.2.0

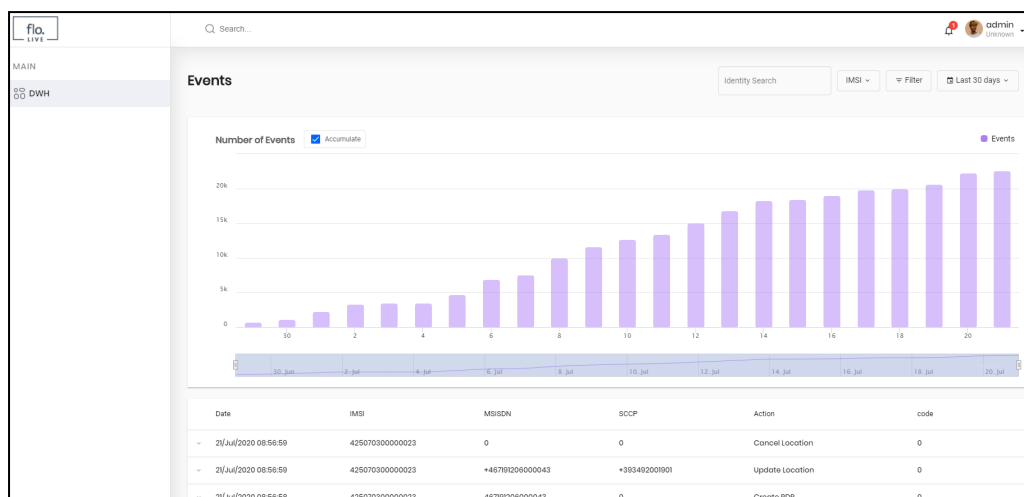
12.1. Signaling Data Warehouse (DWH)

The floLIVE DWH is a centralized and permanent storage location for the data collected from various data sources.

The system features a rich and elaborative device-monitoring module that is part of the entire application suite. This module provides the ability to view and debug all connectivity and service issues of a specific device, such as its historical locations, media attempts (call setup, PDP context), and more.

The floLIVE DWH tool is displayed below:

Figure 10 - DWH Tool



The floLIVE HLR/HSS/UDM supports the sending of multiple signaling events to the DWH. These signaling events may include, among others:

- Send Authentication Info
- Update Location
- Cancel Location
- PSI
- SRI

Each DWH event contains detailed information that describes the specific event, such as APN, charge ID, IMSI, location, etc. The supported formats are CSV, JSON, and ASN1.

12.2. Subscriber Trace

The floLIVE platform has a tool that traces online signaling transactions of a specific device. This tool can invoke an online trace from all floLIVE Core network components regarding a specific device.

The floLIVE UDM supports sending subscriber traces for single or multiple IMSI or MSISDN numbers; the subscriber trace that includes control plane messages for example - DIAMETR.

Information from the UDM is sent to a destination IP and can be monitored or archived by monitoring tools.

EXAMPLE: The following is an example of how to set a subscriber trace from the UDM Admin tool:

Figure 11 - Manage Subscriber Trace

Subscriber Trace

Id	Digits	Destination IP	Actions
RESET			

Rows per page: 20

12.3. Logs

The floLIVE platform supports creating log files for every system network element. The log file includes detailed log messages that are useful for cases of debugging or troubleshooting the system.

Log files can be configured to different debug levels, file sizes, and frequencies.

13. Specifications

13.1. Performance Specifications

The HLR/HSS/UDM can be deployed on a LINUX Containers that work on Kubernetes.

HLR/HSS minimum environment specifications:

Parameter	Specification
Cores	4
Memory	8 GB RAM
Storage	100 GB

Maximum capacity of a single HLR/HSS/UDM node with the above parameters:

Parameter	Specification
Active subscribers	1,000,000
Provisioned (non-active) subscribers	10,000,000
Transactions per second	1000 Includes: • Authentication request • Location updates • Cancel location actions • MT SMS, MO SMS, USSD
API calls per second	200

For a larger capacity, a more powerful VM can be defined or additional VMs can be added.

13.2. Redundancy

For HLR/HSS/UDM geo-redundancy, two or more physical sites are deployed and implements fail over and data replication between them.

The diagram below illustrates the HLR/HSS/UDM Geo Redundancy architecture:

Figure 12 - Geo Redundancy Architecture

